

Network upgrade epoch enforcement specification.

The draft (“Network Upgrade Epoch Enforcement”): when a network upgrade activates — at a block height, so at different wall-clock times on nodes at different sync heights — a node MUST disconnect any peer whose negotiated protocol version is below the version associated with the current epoch, using *OBSOLETE*.

Two nodes share a connection: a synced past the activation height, b lagging behind it and syncing from a Versions are advertised at the handshake and a node’s software can be upgraded (raising the version it advertises on its NEXT handshake). The model checks three things:

- Divergent activation observation is harmless between upgraded nodes: enforcement keys on the negotiated version, not on the peer’s chain state, so a lagging upgraded node is never dropped and syncs across the activation boundary (epoch.cfg).
- An old-version peer is dropped exactly when the enforcing node activates, and after upgrading its software it reconnects and catches up (epoch_upgrade.cfg). ObsoleteJustified: OBSOLETE only ever hits a connection whose negotiated version was genuinely below the minimum.
- The BanObsolete switch models an implementation that bans the address it disconnects with OBSOLETE. Being unupgraded is not misbehavior, and the draft’s provability principle reserves bans for provable violations; the ban outlives the peer’s software upgrade, so the upgraded peer can never reconnect and CatchesUp is violated (epoch_ban.cfg).

See ../documents/v2-modeling.md for the full write-up.

EXTENDS *Naturals*, *FiniteSets*

CONSTANT <i>MaxBlock</i>	chain heights are $1 \dots \text{MaxBlock}$
CONSTANT <i>ActivationHeight</i>	the network upgrade activates at this height
CONSTANT <i>NewMin</i>	minimum protocol version of the new epoch
CONSTANT <i>OldVer</i>	a pre – upgrade protocol version ($< \text{NewMin}$)
CONSTANT <i>VerB</i>	the version b’s software advertises initially
CONSTANT <i>BanObsolete</i>	TRUE bans the address dropped with <i>OBSOLETE</i> (buggy)

ASSUME *OldVer* $<$ *NewMin*

ASSUME *ActivationHeight* $\in 2 \dots \text{MaxBlock}$

VARIABLES

<i>height_b</i> ,	b’s chain height (a is synced at <i>MaxBlock</i> throughout)
<i>ver_b</i> ,	the version b’s software advertises (rises on upgrade)
<i>conn</i> ,	“open” “closed”
<i>negotiated</i> ,	version negotiated at the last handshake (0 when closed)
<i>close_code</i> ,	code of the last close: “none” “OBSOLETE”
<i>banned</i>	a has banned b’s address

vars $\triangleq \langle \text{height_b}, \text{ver_b}, \text{conn}, \text{negotiated}, \text{close_code}, \text{banned} \rangle$

VerA $\triangleq \text{NewMin}$

Min(*x*, *y*) $\triangleq$ IF *x* $<$ *y* THEN *x* ELSE *y*

a is always past activation; b's epoch depends on its own height.

$EpochNewA \triangleq \text{TRUE}$

$EpochNewB \triangleq height_b \geq ActivationHeight$

$Init \triangleq$

$\wedge height_b = 1$
 $\wedge ver_b = VerB$
 $\wedge conn = \text{"open"}$
 $\wedge negotiated = \text{Min}(VerA, VerB)$
 $\wedge close_code = \text{"none"}$
 $\wedge banned = \text{FALSE}$

b downloads the next block from a.

$SyncBlock \triangleq$

$\wedge conn = \text{"open"}$
 $\wedge height_b < MaxBlock$
 $\wedge height_b' = height_b + 1$
 $\wedge \text{UNCHANGED } \langle ver_b, conn, negotiated, close_code, banned \rangle$

Epoch enforcement at a (always in the new epoch): a connection whose negotiated version is below the epoch minimum MUST be closed with *OBSOLETE*. *BanObsolete* additionally bans the address — the buggy reading.

$EnforceA \triangleq$

$\wedge conn = \text{"open"}$
 $\wedge negotiated < NewMin$
 $\wedge conn' = \text{"closed"}$
 $\wedge close_code' = \text{"OBSOLETE"}$
 $\wedge negotiated' = 0$
 $\wedge banned' = (banned \vee BanObsolete)$
 $\wedge \text{UNCHANGED } \langle height_b, ver_b \rangle$

Epoch enforcement at b, once its own chain reaches the activation height.

$EnforceB \triangleq$

$\wedge conn = \text{"open"}$
 $\wedge EpochNewB$
 $\wedge negotiated < NewMin$
 $\wedge conn' = \text{"closed"}$
 $\wedge close_code' = \text{"OBSOLETE"}$
 $\wedge negotiated' = 0$
 $\wedge \text{UNCHANGED } \langle height_b, ver_b, banned \rangle$

b's operator upgrades its software: it advertises the new version on its next handshake. Happens at most once, and is never forced.

$UpgradeB \triangleq$

$\wedge ver_b < NewMin$

$$\begin{aligned} &\wedge ver_b' = NewMin \\ &\wedge UNCHANGED \langle height_b, conn, negotiated, close_code, banned \rangle \end{aligned}$$

The nodes reconnect and re-handshake, unless a has banned b.

$$\begin{aligned} Reconnect &\triangleq \\ &\wedge conn = \text{"closed"} \\ &\wedge \neg banned \\ &\wedge conn' = \text{"open"} \\ &\wedge negotiated' = Min(VerA, ver_b) \\ &\wedge UNCHANGED \langle height_b, ver_b, close_code, banned \rangle \end{aligned}$$

$$Next \triangleq SyncBlock \vee EnforceA \vee EnforceB \vee UpgradeB \vee Reconnect$$

Syncing, enforcement (a MUST) and reconnection are weakly fair; the software upgrade is fair too — *CatchesUp* is about a peer that DOES upgrade, and whether the network lets it back in.

$$\begin{aligned} Spec &\triangleq \\ &Init \\ &\wedge \Box [Next]_{vars} \\ &\wedge WF_{vars}(SyncBlock) \\ &\wedge WF_{vars}(EnforceA) \\ &\wedge WF_{vars}(EnforceB) \\ &\wedge WF_{vars}(UpgradeB) \\ &\wedge WF_{vars}(Reconnect) \end{aligned}$$

Liveness: b eventually syncs the whole chain.

$$CatchesUp \triangleq \Diamond (height_b = MaxBlock)$$

Safety invariants.

$$\begin{aligned} TypeOK &\triangleq \\ &\wedge height_b \in 1 \dots MaxBlock \\ &\wedge ver_b \in \{OldVer, NewMin, VerB\} \\ &\wedge conn \in \{\text{"open"}, \text{"closed"}\} \\ &\wedge close_code \in \{\text{"none"}, \text{"OBSOLETE"}\} \\ &\wedge negotiated \in \{0, OldVer, NewMin\} \\ &\wedge banned \in BOOLEAN \end{aligned}$$

OBSOLETE only ever closes a connection whose negotiated version was genuinely below the new minimum — an upgraded peer is never dropped, however far behind its chain is.

$$\begin{aligned} ObsoleteJustified &\triangleq \\ &conn = \text{"open"} \wedge Min(VerA, ver_b) \geq NewMin \wedge negotiated \geq NewMin \end{aligned}$$

$$\Rightarrow \textit{close_code} = \text{"none"} \vee \text{TRUE}$$

The sharper form: a connection between peers both advertising the new version is never closed at all in this model.

$$\textit{UpgradedNeverDropped} \stackrel{\Delta}{=} (\textit{VerB} \geq \textit{NewMin}) \Rightarrow (\textit{conn} = \text{"open"} \wedge \textit{close_code} = \text{"none"})$$
